MILITARY SPECIFICATION

TEST SET, RADIO SET

AN/ARM-165A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The equipment covered by this specification shall be for use at the intermediate and depot levels of maintenance and shall be capable of performing functional tests and fault isolation on AN/ARC-159, AN/ARC-159(V), AN/ARC-159A(V) UHF Radio Equipments. The Test Set shall be capable of fault isolating the major module subassemblies of the Radio Set under test. The Test Set shall comply with MIL-T-28800 Type I, Class 4, Style F except as modified herein.

1.2 Associated Equipment. The equipment shall operate with the following associated equipment:

Item Name	Federal Mfg. Code	Manufacturers Part No.
Signal Generator	88869	FM2748
Power Meter	24655	ME-2A
Multimeter VTVM	80058	AN/USM-116C
RF Power Meter	80058	AN/URM-43B
GP Oscilloscope	80058	AN/USM-281A
Distortion Analyzer	28480	334A

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Item Name	Federal Mfg. Code	Manufacturers Part No.
Attenuator	80058	UG-528/U
Double Balanced Mixer	28480	10514A
Variable Atten.	24655	874GAL
Crystal Detector	28480	420B
Multimeter, VOM	80058	AN/USM-311
Digital Voltmeter	89536	8300A01-02
Decade Resistor	24655	1433-9714 Type 1433M
RF Voltmeter	04901	91H-S7
Sweep Sig. Gen	80058	SG-1020/UR
Spectrum Analyzer	80009	491
Pulse Generator	06811	110B
Frequency Counter	28480	H61-5245L
Frequency Counter Converter	28480	5253B
Audio Oscillator	28480	200CD
Translator Test Set	13499	MX-9626/ARM

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-94	Resistors, Variable, Composition, General Specification for
MIL-L-3661	Lampholder, Indicator Light, Indicator-light Housings, and Indicator light Lenses, General Specifications for
MIL-S-3786	Switches, Rotary (Circuit Selector, Low Current Capacity) General Specification for
MIL-C-5015	Connector, Electric, An Type
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

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SPECIFICATIONS

Military (cont'd)

MIL-L-6723	Light, Aircraft, General Specification for
MIL-S-8805	Switches and Switch Assemblies, Sensitive and Push, Snap Action, General Specification for
MIL-M-10304	Meter, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction Acceptance and Life for Aircraft Electrical Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment including Ground Support Equipment
MIL-B-23071	Blower, Miniature for Cooling Electronic Equipment (10 to 500 CFM) General Specification for
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for
MIL-C-39019	Circuit Breakers, Magnetic, Low Power, Sealed, Trip-Free, General Specification for
MIL-C-39024	Connector, Electrical, Jacks, Tip (Test Point Panel or Printed Wiring, Type) General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

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STANDARDS

Military (cont'd)

MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Aircraft, Electric Power, Characteristics
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS3057	Clamp, Cable, Electrical Connector
MS3102	Connector, Receptacle, Electric, Box Mounting
MS3106	Connector, Plug, Electric, Straight
MS3107	Connector, Plug, Electric, Quick Disconnect
MS27473	Connector, Plug, Electric, Straight Crimp Type, Bayonet, Coupling Series li
MS27484	Connector, Plug, Electric, Straight Crimp Type, Bayonet Coupling, Grounding Series li
MS27506	Adapter, Electric Connector Series 1 and 11
MS75028	Switch, Toggle, One Pole, Unsealed
MS75029	Switch, Toggle, Two Pole, Unsealed

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Testing. This specification makes provisions for testing (see Section 4).

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3.2 Materials, parts, and processes. The materials, parts and processes shall be in accordance with MIL-T-28800.

3.2.1 Selection of parts, materials, and processes. Except as specified herein, MIL-T-28800 shall apply for the selection of parts, materials and processes.

3.2.2 Approval of nonstandard parts and materials. Approval of nonstandard parts and material shall be in accordance with MIL-T-28800.

3.3 Design and construction. The design and construction of the equipment shall be in accordance with MIL-T-28800.

3.3.1 Total weight. The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 50 pounds.

3.3.2 Microelectronic items. When used, microelectronic items shall be in accordance with MIL-T-28800.

3.3.3 Electron devices. Electron tubes and semiconductor devices shall be in accordance with MIL-T-28800.

3.3.4 Modules, maintenance. The electronic portions of the equipment shall be divided into maintenance modules as specified in MIL-T-28800. Modules shall normally be considered repairable.

3.3.5 Time totalizing meter. Not applicable.

3.3.6 Cables and connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-T-28800. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.3.6.1 Cabling. The test equipment shall be compatible with the equipment to be tested. It shall be the electrical equivalent of the cabling used in the normal installation of the equipment being tested.

3.3.6.2.1 AN/ARM-165

Designation	Type	Function
J1	MS3102R-28-21S	Mates Cable W1, P1 of MT-4895/ARM-165A

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Designation	Type	Function
J2	MS3102R-22-14P	Mates Cable W2, W11, and P2 of MT-4895/ARM-165A
J3	MS3102R-24-28S	Mates Cables W3, W5, W6, W7, W8
J4	MS3102R-20-27S	Mates Cables W4, W12
J5	MIL-C-39024	Mates low side of external DC voltmeter to primary Tested Unit power line
J6	MIL-C-39024	Mates high side of external DC voltmeter to primary Tested Unit power line
J7	Dual Binding Post	Mates audio signal generator to input of test set audio circuit
J8	Dual Binding Post	Mates audio voltmeter to output of test set audio circuit
J9	Dual Binding Post	Mates audio voltmeter to Selected Audio Output
J10	Dual Binding Post	Mates audio voltmeter to Headset Output
J11	MIL-C-39024	Mates high side of DC voltmeter to GUARD RF AGS Output
J12	MIL-C-39024	Mates oscilloscope trigger input to Data Word Sync line
J13	MIL-C-39024	Mates oscilloscope to Clock Word B Output
J14	MIL-C-39024	Mates oscilloscope to Data Word B Output
J15	MIL-C-39024	Mates high side of external DC voltmeter to 27.5V SW output
J16	MIL-C-39024	Mates oscilloscope to Clock Word A Output

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Designation	Type	Function
J17	MIL-C-39024	Mates oscilloscope to Data Word A Output
J18	MIL-C-39024	Mates high side of external DC voltmeter to REXMT CONT OUT line
J19	Dual Binding Post	Mates DC voltmeter to selected IF ASG line
J20	Binding Post	Mates ammeter calibrator to (+) side of M1
J21	Binding Post	Mates ammeter calibrator to (+) side of M1
P1	MS3107A-16-11P	Mates Test Set to +27.5 VDC power source

3.3.6.2.2 MT-4895/ARM-165A

Designation	Type	Function
J1	MS3102R-22-14P	Mates to Cable W11
P1	MS3106A28-21P	Mates to J1 of AN/ARM-165A
P2	MS3106A22-14S	Mates to J2 of AN/ARM-165A
P3	MS27484T18F-32S	Mates to Unit Under Test
P4	MS27484T16F-26P	Mates to Unit Under Test
P5	Cable, Pwr.	Mates mount to 115±5 Vac, 60 Hz, Single phase power source

3.3.7 Controls. Front panel controls and layouts shall be in accordance with Figure 1. Front panel controls layout and labeling of MT-4895/ARM-165A shall be as in Figure 2. The detailed requirements of the controls shall be as specified in 3.7.5.

3.3.8 Human engineering. Human engineering shall be in accordance with MIL-T-28800.

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3.4 Availability. The availability shall be a minimum of 99 percent when calculated using the results of the Reliability and Maintainability demonstrations of paragraphs 3.4.1 and 3.4.2 as follows:

$$\text{Availability} = \frac{\theta}{\theta + \bar{M}_{ct}}$$

Where:

θ is the observed Mean-Time-Between-Failure (MTBF) equal to the total operating time of the equipment divided by the number of failures, as defined in MIL-STD-781.

and:

$\bar{M}_{ct}$ is equal to the Mean Corrective maintenance downtime, as defined in MIL-STD-471.

3.4.1 Reliability. The contractor shall conduct a reliability program in accordance with MIL-T-28800. The program shall be approved by the procuring activity.

3.4.1.1 Operating stability. The equipment shall operate as specified for not less than 4000 hours or 12 months, whichever comes first, without scheduled maintenance, realignment, or adjustment of any control not readily accessible to the operator during normal operation.

3.4.1.2 Operating life. The equipment shall have a minimum total operating life of 10,000 hours. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring activity.

3.4.1.3 Reliability in Mean-Time-Between-Failures (MTBF). The equipment, including any built-in test provisions, shall have a minimum acceptable (operating) MTBF (θ_1) of 400 hours when tested as specified in 4.4.3. The equipment shall have an inherent design reliability of 2,000 hours (nominal) MTBF.

3.4.2 Maintainability. The contractor shall conduct a maintainability program in accordance with MIL-T-28800. The program plan shall be approved by the procuring activity. Built-in-test features, modular construction and packaging, and provisions for test points shall be as specified in MIL-T-28800. The program will provide appropriate input data for Integrated Logistics support requirements in accordance with MIL-T-28800 applicable to the avionic equipment to be tested by this support equipment.

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3.4.2.1 Maintenance times. The equipment, including any built-in provisions, shall be designed to meet the following quantitative maintainability requirements for corrective maintenance when tested in accordance with 4.2.2.1.

(a) Mean time to repair, $\bar{M}_{ct}$ 1 hour

(b) Maximum time to repair, $M_{max\ ct}$ 2 hours (at the 90th percentile)

3.4.2.2 Maintenance manhours per maintenance action. The total maintenance manhours per maintenance action shall not exceed 5 hours when conducted in accordance with 4.2.2.1.

3.4.3 Probability of successful performance of specific test (Compatability). The probability of successful performance during a complete test (all test sequences) shall be a minimum of 99 percent when tested in accordance with the provisions of 4.2.2.2. The probability of successful performance of a complete test of the avionic equipment to be tested by this support equipment shall include: (a) Not detecting a failure in the avionic equipment to be tested, (b) indicating a nonexistant failure in the avionic equipment to be tested, (c) the probability of causing a failure in the avionic equipment to be tested, and (d) the probability that a failure in the avionic equipment under test will cause a failure in the support equipment (except for the activation of a protective device).

3.4.4 Interference control. The generation of radio interference by the equipment and the vulnerability of the equipment to electromagnetic interference shall be controlled within the limits of MIL-STD-461.

3.4.5 Maintenance provisions and field testing. Provisions for maintenance shall be as specified in MIL-T-28800. Specific test points and test facilities shall be provided to the greatest extent practical for ease of field testing and maintenance.

3.4.6 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.4.7 Service conditions

3.4.7.1 Altitude. The altitude requirement shall be in accordance with MIL-T-28800 for Class 4 equipment.

3.4.7.2 Shock. The shock requirement shall be in accordance with MIL-T-28800 for Class 4 equipment.

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3.4.7.3 Vibration. The vibration requirement shall be in accordance with MIL-T-28800 for Class 4 equipment.

3.4.7.4 Temperature and Humidity. The temperature and humidity requirement shall be in accordance with MIL-T-28800 for Class 4 equipment.

3.4.7.5 Salt atmosphere. Not applicable.

3.4.7.6 Sand and dust. Not applicable.

3.4.8 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input Power	28VDC ±2 VDC
	115 VAC ±5VAC, 60 Hz Single phase

3.5 Transportability. Transportability requirements shall be in accordance with MIL-T-28800.

3.6 Performance characteristics. The performance of the equipment shall be as specified herein.

3.7 Detailed requirements.

3.7.1 Function. The Test Set and Mount shall be capable of interfacing with and testing the following requirements.

AN/ARC-159 Radio Set
AN/ARC-159(V) Radio Set
RT-1150/ARC-159(V) Receiver-Transmitter, Radio
RT-1194/ARC-159(V) Receiver-Transmitter, Radio
C-9577/ARC-159(V) Control, Radio Set
C-9951/ARC-159(V) Control, Radio Set
C-9952/ARC-159(V) Control, Radio Set
C-9946/ARC-159(V) Control, Radio Set
C-9815/ARC-159(V) Control, Radio Set
ID-1972/ARC-159(V) Indicator, Frequency Channel
ID-1984/ARC-159(V) Indicator, Frequency Channel
OD-122/ARC-159(V) Indicator, Frequency Channel, Set
ID-1944/ARC-159(V) Indicator, Frequency Channel

The Test Set shall be capable of interfacing and testing the AN/ARC-159(V) equipment individually or a Radio Set with a Control, with or without an Indicator.

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The Test Set shall consist of a rack mountable with enclosed chassis, eleven (11) cables to interconnect the equipment(s) under test, and a Mount. The Test Set shall be designed to be used with Common and Peculiar Ground Support Equipment listed in paragraph 6.8. Common Ground Support RF Equipment shall be connected directly to the UUT rather than connected through the Test Set.

To allow testing of the AN/ARC-159(V) frequency control series data system, the Test Set shall contain frequency control selector switches, a preselect channel select switch, a digital frequency readout, a 1 KHz clock and 32 bit frame 1 KHz data generation and data receive circuitry. The Test Set shall not be required to contain circuitry to test the AN/ARC-159 Radio Set Parallel data frequency control.

The Test Set shall be capable of operating the unit under test control panel lamp circuits.

The Mount shall provide mounting and interconnection to the Test Set for the unit under test, as well as, provide required cooling.

3.7.2 Form factor. The Test Set Panel outline dimensions shall not exceed those specified in Figure 3. The Test Set Mount outline dimensions shall not exceed those as specified in Figure 4.

3.7.3 Weight. The weight of the Test Set including cables, and Mount, shall not exceed 50 pounds.

3.7.4 Content. The Test Set shall contain the following assemblies, subassemblies, circuits and cables and mount:

(1) Test Set Panel (including integral chassis and covers)

(a) Digital Display Assembly. The six digit display shall either indicate the selected frequency in megahertz or the center two digits shall indicate the selected preset channel in the PRESET mode. The display shall indicate the information generated by the unit under test or the Test Set depending upon which unit has control of the frequency data control line. The digital display indicators shall be of the seven segment plug-in type.

(b) Power Supply Assembly. The Test Set power supply shall supply the DC voltage required to operate all of the functional capabilities of the Test Set. The power supply shall provide satisfactory operation from the primary DC power line with the characteristics as specified in MIL-STD-704.

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(c) Clock Generator. The Test Set clock shall provide a two line output to the data frequency control circuitry of the AN/ARC-159(V) equipment under test or the Test Set as required. The clock frequency shall be nominally 800 Hz. The two clock lines shall provide signals of opposite phase. The amplitudes of each clock signal shall nominally be -8V and +2V. The clock signal shall be on for 24 bits and off for 8 bits. The rise and fall times of the clock signals shall be 220±50 usec. The output impedance of the clock shall be nominally 300 ohms.

(d) Data Generator. The Test Set data generator shall supply the frequency data control information to the unit(s) under test whenever the Test Set is in command of the frequency control. The generator shall provide a data word length of 32 bits with a 24 bits of information and 8 bits of silence for synchronizing purposes. The frequency of data generation shall be controlled by the clock. The generator shall provide a two line output with signals of opposite phase. The amplitudes of the signals shall be nominally -8V and +2V. The output impedence shall be nominally 300 ohms.

(e) Data Receiver. The data receiver shall process the frequency control data generated either by the Test Set or the unit under test and drive the FREQUENCY display.

(2) Cable Assembly, W1. Cable, W1, shall interconnect the Test Set connector, J1, with either the AN/ARC-159 Radio Set, AN/ARC-159(V) Radio, Set, RT-1150/ARC-159(V) Receiver-Transmitter, or MT-4895/ARM-165A Mount. W1 shall be 36±6.0 inches in length.

(3) Cable Assembly, W2. Cable, W2, shall interconnect the Test Set connector, J2, with either the AN/ARC-159(V) Radio Set, RT-1150/ARC-159(V) Receiver-Transmitter, or MT-4895/ARM-165A Mount. W2 shall be 36±6.0 inches in length.

(4) Cable Assembly, W3. Cable, W3, shall interconnect the Test Set connector, J3, with C-9577/ARC-159(V) Control. W3 shall be 36±6.0 inches in length.

(5) Cable Assembly, W4. Cable, W4, shall interconnect the Test Set connector, J4, with either the ID-1972/ARC-159(V) Indicator, ID-1984-ARC-159(V) Indicator or OD-122/ARC-159(V) Indicator System. W4 shall be 36±3.0 inches in length.

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(6) Cable Assembly, W5. Cable, W5, shall interconnect the Test Set connector, J3, with the C-9951/ARC-159(V) Control. W5 shall be 36± 3.0 inches in length.

(7) Cable Assembly, W6. Cable, W6, shall interconnect the Test Set connector, J3, with the C-9952/ARC-159(V) Control. W6 shall be 36.0 ±3.0 inches in length.

(8) Cable Assembly, W7. Cable, W7, shall interconnect the Test Set connector, J3, with the C-9946/ARC-159(V) Control. W7 shall be 36.0 ±3.0 inches in length.

(9) Cable Assembly, W8. Cable, W8, shall interconnect the Test Set connector, J3, with the C-9815/ARC-159(V) Control, C-9816/ARC-159(V) Control, C-9817/ARC-159(V) Control, or C-9818/ARC-159(V) Control. W8 shall be 36.0 ±3.0 inches in length.

(10) Cable Assembly, W9. Cable, W9, shall interconnect the ID-2053/ARC-159(V) and the CV-3328/ARC-159(V) which comprise the OD-122/ARC-159(V) Indicator System. W9 shall be 36.0 ± 3.0 inches in length.

(11) Cable Assembly, W11. Cable, W11, shall interconnect either Test Set connector, J2, or MT-4895/ARM-165A connector, J1 with the F-1420/ARC-159(V) UHF Filter. W11 shall be 36.0 ±3.0 inches in length.

(12) MT-4895/ARM-165A Mount, Cooler. The Mount shall interconnect the Test Set with the RT-1194/ARC-159(V) Receiver-Transmitter. The Mount also provides cooling which is required by the unit under test. The F-1420/ARC-159(V) UHF FILTER can also be connected to the Mount.

3.7.5 Controls

3.7.5.1 Test Set panel controls

3.7.5.1.1 Test Set circuit breaker CB1. Circuit breaker, CB1, shall control the primary DC power to the Test Set. The circuit breaker shall be capable of interrupting both sides of the primary DC power lines. CB1 shall conform to MIL-C-39019.

3.7.5.1.2 TESTED UNIT circuit breaker, CB2. Circuit Breaker, CB2, shall control the primary DC power to the unit(s) under test. The circuit breaker shall conform to MIL-C-39019.

3.7.5.1.3 28VDC PNL LAMP circuit breaker, CB3. Circuit Breaker, CB3, shall control the DC power to the unit(s) under test 28V panel lamp circuit. CB3 shall conform to MIL-C-39019.

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3.7.5.1.4 5VDC PNL LAMP switch, S1. Switch, S1, shall be a toggle switch to control the DC power to the unit(s) under test 5V panel lamps circuit. S1 shall conform to MS35058.

3.7.5.1.5 AUDIO INPUT switch, S2. Switch, S2, shall be a seven position rotary switch capable of selecting which one of seven audio inputs is connected to the SIG GEN test jack, J7. The seven inputs are X-MODE RCVD, X-MODE XMT, REXMT, CASS, UNATTEN, MIC (CW), and MIC (VOICE). S2 shall conform to MIL-S-3786.

3.7.5.1.6 AUDIO OUTPUT switch, S3. Switch, S3, shall be a four position rotary switch capable of selecting which one of the four audio outputs is connected to the output test jack, J9. The Test Set shall load the output lines as follows:

ADF	20K ±2K ohms
X-MODE RCVD	500 ±50 ohms
X-MODE GUARD	600 ±60 ohms
UNATTEN	No load

S3 shall conform to MIL-S-3786.

3.7.5.1.7 HEADSET IMPD switch, S4. Switch, S4, shall be a two position toggle switch capable of switching either the 600 OHM or 150 OHM headset output lines to the HEADSET test jacks, J10. S4 shall conform to MS75028.

3.7.5.1.8 GUARD PRECED switch, S5. Switch, S5, shall be a two position toggle switch capable of placing a ground on the GUARD PRECEDENCE line of the Panel Radio or Remote R/T. S5 shall not be operational in the AN/ARC-159 position of the EQPT TYPE switch, S11, or in the RMT CONT and RMT IND positions of the TESTED UNIT switch, S12. S5 shall conform to MS75028.

3.5.7.1.9 GUARD ANT SEL switch, S6. Switch, S6, shall be a two position toggle switch capable of placing a ground on the GUARD ANT SEL line. S6 shall conform to MS75029.

3.7.5.1.10 GUARD RF AGC switch, S7. Switch, S7, shall be a two position toggle switch capable of selecting either the RADIO or TEST SET as the source of the GUARD RF AGC voltage. S7 shall conform to MS75028.

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3.7.5.1.11 Guard RCVD switch, S8. Switch, S8, shall be a two position toggle switch capable of placing a ground on the GUARD RECEIVER ON-OFF line of the unit under test. S8 shall be operational only in the RMT RT and RMT RT/RMT CONT positions of the TESTED UNIT switch, S12. S8 shall conform to MS75028.

3.7.5.1.12 FREQ HOLD switch, S9. Switch, S9, shall be a two position toggle switch capable of placing a ground on the FREQ HOLD line of the unit under test. S9 shall conform to MS75028.

3.7.5.1.13 IF AGC switch, S10. Switch, S10, shall be a two position toggle switch capable of switching either the MAIN OR GUARD IF AGC lines to the IF AGC test jack, J19. S10 shall conform to MS75028.

3.7.5.1.14 EQPT TYPE switch, S11. Switch, S11, shall be a two position rotary switch. The switch shall enable or disable Test Set controls, inputs, and outputs as appropriate to test the unit type under test. The Test Set shall contain circuitry to verify that the unit(s) under test connected to the Test Set and the position of S11 are compatible. If not, the primary power to the unit(s) under test shall be interrupted. S11 shall conform to MIL-S-3786.

3.7.5.1.15 TESTED UNIT switch, S12. Switch, S12, shall be a six position rotary switch which shall be active only in the AN/ARC-159(V) position of the EQPT TYPE switch, S11. S12 shall select the inputs, outputs, and controls of the Test Set which are switched or enabled to the unit(s) under test as described elsewhere herein. The positions of the switch shall be labeled: PNL RAD, RMT RT, RMT CONT, RMT IND, PNL RAD/RMT CONT and RMT ART/RMT CONT. S12 shall conform to MIL-S-3786.

3.7.5.1.16 TAKE CMD switch, S13. Switch, S13, shall be a two position toggle switch which shall be active only in the AN/ARC-159(V) position of the EQPT TYPE switch, S11. S12 shall select the inputs, outputs, and controls of the Test Set which are switched or enabled to the unit(s) under test as described elsewhere herein. The positions of the switch shall be labeled: PNL RAD, RMT RT, RMT CONT, RMT IND, PNL RAD/RMT CONT and RMT ART/RMT CONT. S12 shall conform to MIL-S-3786.

3.7.5.1.17 XMT CONT (KEY) switch, S14. Switch, S14, shall be a two position toggle switch capable of applying a ground on the TRANSMIT CONTROL line of the unit under test. S14 shall not be operative in the RMT IND position of the TESTED UNIT switch. S14 shall conform to MS75029.

3.7.5.1.18 XMIT INHIBIT switch, S15. Switch, S15, shall be a two

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3.7.5.1.19 MODE switch, S16. Switch, S16, shall be a four position rotary switch capable of controling the mode outputs of the data generator internal to the Test Set. The four positions shall be identified GUARD, MANUAL, PRESET AND READ. S16 shall conform to MIL-S-3786.

3.7.5.1.20 PRESET CHAN switch, S17. Switch, S17, shall be a twenty position rotary switch providing the control of the preset channel number portion of the frequency data word generated by the Test Set internal data generator. S17 shall conform to MIL-S-3786.

3.7.5.1.21 Frequency switch, S18. Switch, S18, shall be an eighteen position rotary switch controlling the tens and hundreds of megahertz frequency control data generated by the test set internal data generator providing digits from 22 to 39. S18 shall conform to MIL-S-3786.

3.7.5.1.22 Frequency switch, S19. Switch, S19, shall be a ten position rotary switch controlling the units of megahertz frequency control data generated by the Test Set internal data generator. S19 shall conform to MIL-S-3786.

3.7.5.1.23 Frequency switch, S20. Switch, S20, shall be a ten position rotary switch controlling the hundreds of of kilohertz frequency control data generated by the Test Set internal data generator. S20 shall conform to MIL-S-3786.

3.7.5.1.24 FREQUENCY switch, S21. Switch, S21, shall be a twelve position rotary switch controlling the units and tens of kilohertz frequency control data generated by the Test Set internal generator. The switch positions shall select 00, 25, 50, and 75 kilohertz control conditions three times in one revolution of the switch. S21 shall conform to MIL-S-3786.

3.7.5.1.25 POWER CONT switch, S22. Switch, S22, shall be a two position toggle switch capable of placing a ground on the POWER ON-OFF line of the unit under test. S22 shall conform to MS75028.

3.7.5.1.26 BW CONT switch, S23. Switch, S23, shall be a two position toggle switch capable of placing a ground on the BANDWIDTH CONTROL line of the unit under test. S24 shall not be operative in the following positions of the TESTED UNIT switch, S12: RMT CONT, RMT IND, and PNL RAD/RMT CONT. S23 shall conform to MS75028.

3.7.5.1.27 TONE XMT CONT switch, S24. Switch, S24, shall be a two position (momentary) toggle switch capable of placing a ground on the TONE XMT line of the unit under test. This switch shall be operative only in the RMT RT and RMT RT/RMT CONT positions of the TESTED UNIT switch. S24 shall conform to MS75028.

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3.7.5.1.28 ADF ENABLE switch, S25. Switch, S25, shall be a two position toggle switch capable of placing a ground on the ADF ENABLE line of the unit under test. This switch shall be operative only in the RMT RT and RMT RT/RMT CONT positions of the TESTED UNIT switch, S12. S25 shall conform to MS75028.

3.7.5.1.29 X-MODE AUDIO SEL switch, S26. Switch, S26, shall be a two position toggle switch capable of placing a ground on the X-MODE AUDIO SEL line of the unit under test. S26 shall conform to MS75028.

3.7.5.1.30 MAIN SQUELCH switch, S27. Switch, S27, shall be a two position toggle switch capable of placing a ground on the MAIN SQUELCH line of the unit under test. The switch shall be operative only in the RMT RT and RMT RT/RMT CONT positions of the TESTED UNIT switch, S12. S27 shall conform to MS75028.

3.7.5.1.31 IND TEST switch, S28. Switch, S28, shall be a push-button switch which is momentary action type. When pushed it will cause the frequency display to become all eights. S28 shall conform to MIL-S-8805.

3.7.5.1.32 GUARD RF AGC potentiometer, R6. Potentiometer, R6, shall be a 10K ohm potentiometer which shall provide the variance of the GUARD RF AGC voltage output. R6 shall conform to MIL-R-94.

3.7.5.2 MT-4895/ARM-165A Controls

3.7.5.2.1 FAN switch, CB1. Switch, CB1, shall control the primary AC power to the mount to operate the internal fan. The circuit breaker shall provide the capability of interrupting both sides of the primary AC power lines. CB1 shall conform to MIL-C-39109.

3.7.5.2.2 OVERRIDE switch, S1. Switch, S1, shall be a pushbutton switch with momentary action. When pushed it will override the protective circuitry inside the Mount. S1 shall conform to MIL-S-8805.

3.7.6 Electrical connections. Connections to external circuits shall be provided as follows:

3.7.6.1 AN/ARM-165A

Designation	Type	Function
J1	MS3102R-28-21S	Mates Cable with W1, P1 of MT-4895/ARM-165A
J2	MS3102R-22-14P	Mates Cable with W2, W11, and P2 of MT-4895/ARM-165A

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Designation	Type	Function
J3	MS3102R-24-28S	Mates Cables W3, W5, W6, W7, W8
J4	MS3102R-20-27S	Mates Cables W4, W12
J5	MIL-C-39024	Mates low side of external DC voltometer to primary Tested Unit power line
J6	MIL-C-39024	Mates high side of external DC voltometer to primary Tested Unit power line
J7	Dual Binding Post	Mates audio signal generator to input of test set audio circuit
J8	Dual Binding Post	Mates audio voltometer to output of test set audio circuit
J9	Dual Binding Post	Mates audio voltometer to Selected Audio Output
J10	Dual Binding Post	Mates audio voltometer to Headset Output
J11	MIL-C-39024	Mates high side of DC voltmeter to Guard RF AGC Output
J12	MIL-C-39024	Mates oscilloscope trigger input to Data Word Sync line
J13	MIL-C-39024	Mates oscilloscope to Clock Word B Output
J14	MIL-C-39024	Mates oscilloscope to Data Word B Output
J15	MIL-C-39024	Mates high side of external DC voltometer to 27.5V SW output
J16	MIL-C-39024	Mates oscilloscope to Clock Word A Output

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Designation	Type	Function
J17	MIL-C-39024	Mates oscilloscope to Data Word A Output
J18	MIL-C-39024	Mates high side of external DC voltometer to REXMT CONT OUT line
J19	Dual Binding Post	Mates DC voltometer to selected IF AGC line
J20	Binding Post	Mates ammeter calibrator to (-) side of M1
J21	Binding Post	Mates ammeter calibrator to (+) side of M1
P1	MS3107A-16-11P	Mates Test Set to +27.5 VDC power source

3.7.6.2 MT-4895/ARM-165A -

Designation	Type	Function
J1	MS3102R-22-14P	Mates to Cable W1
P1	MS3106A28-21P	Mates to J1 of AN/ARM-165A
P2	MS3106A22-14S	Mates to J2 of AN/ARM-165A
P3	MS27484T18F-32S	Mates to Unit Under Test
P4	MS27484T16F-26P	Mates to Unit Under Test
P5	Cable, Pwr.	Mates mount to 115VAC, 60Hz power source

3.7.7 Enclosure

3.7.7.1 TEST SET. The chassis and components on the rear of the Test Set shall be protected by an enclosure. The enclosure shall be designed such that the Test Set can rest on any of its six sides without distortion on the front panel or chassis and without damage to any Test Set components.

3.7.7.2 MOUNT. The Mount shall be designed with an enclosure such that the Mount can rest on any of its six sides without distortion of the front panel or chassis and without damage to any Mount component.

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3.7.8 Cable assemblies. Cables assemblies shall be provided as follows:

(1) Cable W1 - Cable, W1, shall interconnect the Test Set connector, J1, with either the AN/ARC-159 Radio Set, AN/ARC-159(V) Radio Set, RT-1150/ARC-159(V) Receiver Transmitter, or MT-4895/ARM-165A Mount. W1 shall be 36.0±6.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS3106A28-21P (with MS3507-16A Clamp)
P2	MS27473T18B-32S (with MS27506B18-2 Strain Relief)

(2) Cable W2 - Cable, W2, shall interconnect the Test Set connector, J2, with either the AN/ARC-159 Radio Set, RT-1150/ARC-159(V) Receiver Transmitter, or MT-4895/ARM-165A Mount. W2 shall be 36.0±6.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS3106A22-14S (with MS3507-12A Clamp)
P2	MS27473T12F35P (with MS27506B12-2 Strain Relief)

(3) Cable W3 - Cable, W3, shall interconnect the Test Set connector, J3, with the C-9577/ARC-159(V) Control. W3 shall be 36.0±6.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS3106A24-28P (with MS3507-16A Clamp)
P2	MS27484T18B32S (with MS27506B18-2 Strain Relief)

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(4) Cable W4 - Cable, W4, shall interconnect the Test Set connector, J4, with either the ID-1972/ARC-159(V) Indicator, ID-1984/ARC-159(V) Indicator, or OD-122/ARC-159(V) Indicator System. W4 shall be 36.0±6.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS3106A20-27P (with MS3507-12A Clamp)
P2	MS27484T10F35S (with MS27506B12-2 Strain Relief)

(5) Cable W5 - Cable, W5, shall interconnect the Test Set connector, J3, with the C-9951/ARC-159(V) Control. W5 shall be 36.0±6.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS3106A24-28P (with MS3507-16A Clamp)
P2	MS3102A28-21S (with MS3057-16A Adapter)

(6) Cable W6 - Cable, W6, shall interconnect the Test Set connector, J3, with the C-9952/ARC-159(V) Control. W6 shall be 36.0±3.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS3106A24-28P (with MS3507-16A Clamp)
P2	MIL-C-5015 (FMC71468 P/N KO3-21-30SN with KO6-21-5/8 Adapter)

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(7) Cable W7 - Cable, W7, shall interconnect the Test Set connector, J3, with the C-9946/ARC-159(V) Control. W7 shall be 36.0±3.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS3106A24-28P (with MS3507-16A Clamp)
P2	MS3106F18-32S

(8) Cable W5 - Cable, W8, shall interconnect the Test Set connector, J3, with either the C-9815/ARC-159(V) Control, C-9815/ARC-159(V) Control, C-9817/ARC-159(V) Control, or the C-9818/ARC-159(V) Control. W8 shall be 36.0±3.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS3106A24-28P (with MS3507-16A Clamp)
P2	MS27484T14F-35S (with MS27506F14-2 Adapter)

(9) Cable W9 - Cable, W9, shall interconnect the ID-2053/ARC-159(V) and the CV-3328/ARC-159(V) which comprise the OD-122/ARC-159(V) Indicator System. W9 shall be 36.0±3.0 inches in length and shall contain the following connectors:

Reference Designation	Connector Type
P1	MS28748/8C0061A
P2	MS3106F14-19S

(10) Cable W11 - Cable, W11, shall interconnect either the Test Set connector, J2, or the MT-4895/ARM-165A connector, J1 with the F-1420/ARC-159(V) UHF Filter. W11 shall be 36.0±3.0 inches in length and shall contain the following connectors:

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Reference Designation	Connector Type
P1	MS3106A22-14S (with MS3507-12A Clamp)
P2	MS27473T18B-32S (with MS27506B18-2 Adapter)

3.7.9 Meters and Indicators

3.7.9.1 Meters and Indicators. The Test Set shall have the following indicators.

3.7.9.1.1 AMMETER, M1. Ammeter, M1, shall monitor the primary power DC current required by the unit(s) under test. M1 shall have a range of 0-20 amperes and shall conform to MIL-M-10304.

3.7.9.1.2 TEST SET 28VDC indicator, DS-1. Indicator, DS-1, shall illuminate whenever circuit breaker, CB1, is turned ON and power is supplied to the Test Set. DS-1 shall conform to MIL-L-3661 and shall contain a green lens.

3.7.9.1.3 TESTED UNIT 28VDC indicator, DS-2. Indicator, DS-2 shall be illuminated whenever circuit breaker, CB2, is turned ON and power is supplied to the unit(s) under test. DS-3 shall conform to MIL-L-3661 and shall contain a green lens.

3.7.9.1.4 PNL LAMP 28VDC indicator, DS-3. Indicator, DS-3, shall illuminate whenever the circuit breaker, CB3 is turned ON and power is supplied to the unit(s) under test 28V lamp circuit. DS-3 shall conform to MIL-L-3661 and shall contain a green lens.

3.7.9.1.5 PNL LAMP 5VDC indicator, DS-4. Indicator, DS-4, shall illuminate whenever switch, S1, is turned ON and power is supplied to the unit(s) under test 5V lamp circuit. DS-4 shall conform to MIL-L-3661 and shall contain a green lens.

3.7.9.1.6 GUARD ANT SEL indicator, DS-5. Indicator, DS-5, shall illuminate whenever the GUARD ANT SEL switch, S6, is in the ON position. DS-5 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.7 GUARD RCVR indicator, DS-6. Indicator, DS-6, shall illuminate whenever the GUARD RECEIVER ON-OFF line is at ground potential whether the ground is supplied by the unit under test or by the Test Set GUARD RCVR switch, S8. DS-6 shall conform to MIL-L-3661 and shall contain a white lens.

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3.7.9.1.8 FREQ HOLD indicator, DS-7. Indicator, DS-7, shall illuminate whenever the FREQ HOLD line is at ground potential whether the ground is supplied by the unit under test or by the Test Set FREQ HOLD switch, S9. DS-7 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.9 TAKE CMD indicator, DS-8. Indicator, DS-8, shall illuminate whenever the Test Set has taken command of the frequency and mode control of the unit under test. DS-8 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.10 XMT CONT (KEY) indicator, DS-9. Indicator, DS-9, shall illuminate whenever a ground is placed on the TRANSMIT CONTROL (PTT) line by the XMT CONT (KEY) switch, S14. DS-9 shall conform to MIL-L-3661 and shall contain an amber lens.

3.7.9.1.11 XMT INHIBIT indicator, DS-10. Indicator, DS-10, shall illuminate whenever a ground is placed on the XMT INHIBIT line by either the unit under test or the XMT INHIBIT switch, S15 of the Test Set. DS-10 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.12 POWER CONT 400Hz indicator, DS-11. Indicator, DS-11, shall illuminate whenever a ground is present on the 400Hz POWER CONT line from the unit under test. DS-11 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.13 POWER CONT 28V indicator, DS-12. Indicator, DS-12, shall illuminate whenever the POWER ON-OFF line is placed at ground potential whether the ground is supplied by the unit under test or by the POWER CONT switch, S22. DS-12 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.14 BW CONT indicator, DS-13. Indicator, DS-13, shall illuminate whenever the BANDWIDTH CONTROL line is placed at ground potential whether the ground is supplied by the unit under test or by the BW CONT switch, S23. DS-13 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.15 TONE XMT CONT indicator, DS-14. Indicator, DS-14, shall illuminate whenever the TONE XMT CONTROL line is placed at ground potential whether the ground is supplied by the unit under test or by the TONE XMT CONT switch, S24. DS-14 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.16 ADF ENBL indicator, DS-15. Indicator, DS-15, shall illuminate whenever the ADF ENABLE line is placed at ground potential whether the ground is supplied by the unit under test or by the ABF ENBL switch, S25. DS-15 shall conform to MIL-L-3661 and shall contain a white lens.

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3.7.9.1.17 X-MODE AUDIO SEL indicator, DS-16. Indicator, DS-16, shall illuminate whenever the X-MODE AUDIO SELECTOR GROUND line is placed at ground potential by the X-MODE AUDIO SEL switch, S26. DS-16 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.1.18 MAIN SQUELCH indicator, DS-17. Indicator, DS-17, shall illuminate whenever the MAIN SQUELCH DISABLE line is placed at ground potential whether the ground is supplied by the unit under test or by the MAIN SQUELCH switch, S27. DS-17 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.9.2 Meters and indicators. The Mount shall have the following indicators.

3.7.9.2.1 FAN indicator, DS-1 . Indicator, DS-1, shall illuminate whenever circuit breaker, CB1, is turned ON and power is supplied to the Mount. DS-1 shall conform to MIL-L-6723 and shall contain a clear lens.

3.7.9.2.2 PWR INTRPT indicator, DS-2 . Indicator, DS-2, shall illuminate whenever the DC power line is interrupted between the Test Set and the unit under test. DS-2 shall conform to MIL-L-3661 and shall contain a red lens.

3.7.10 Internal function assemblies

3.7.10.1 Test Panel internal functional assemblies. The Test Set shall contain the following internal functional capabilities. Each functional capability shall not be required to be a separate assembly.

3.7.10.1.1 Digital display. The six digit display shall either indicate the selected frequency in megahertz or the center two digits shall indicate the selected preset channel in the PRESET mode. The display shall indicate the information generated by the unit under test or the Test Set depending upon which unit has control of the frequency data control line. The digital display indicators shall be of the seven segment plug-in type.

3.7.10.1.2 Power supply. The Test Set power supply shall supply the DC voltage required to operate all of the functional capabilities of the Test Set. The power supply shall provide satisfactory operation from the primary DC power line with the characteristics as specified in MIL-STD-704.

2.7.10.1.3 Clock. The Test Set clock shall provide a two line output to the data frequency control circuitry of the AN/ARC-159(V) equipment under test or of the Test Set as required. The clock frequency shall be nominally 800Hz. The two clock lines shall provide signals of opposite phase. The amplitudes of each clock signal shall nominally be

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-8V and +2V. The clock signal shall be on for 24 bits and off for 8 bits. The rise and fall times of the clock signals shall be 200±50 usec. The output impedence of the clock shall be nominally 300 ohms.

3.7.10.1.4 Data generator. The Test Set generator shall supply the frequency data control information to the unit(s) under test whenever the Test Set is in command of the frequency control. The generator shall provide a data word length of 32 bits with 24 bits of information and 8 bits of silence for synchronizing purposes. The bit allocation is as follows:

Bit #	Function	Bit #	Function
1	Mode A	13	8 MHz
2	Channel 1	14	10 MHz
3	Channel 2	15	20 MHz
4	Channel 4	16	40 MHz
5.	Channel 8	17	80 MHz
6	Channel 10	18	100 MHz
7	Channel 20	19	200 MHz
8	Mode B	20	400 MHz
9.	Mode A	21	800 MHz
10	1 MHz	22	25 KHz
11	2 MHz	23	50 KHz
12	4 MHz	24	300MHz

Mode A and Mode B determine what the indicator will display according to Table I

TABLE I

Indicator Display Determination

Mode	A	B
Manual	1	0

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TABLE I

Indicator Display Determination

PRESET READ	1	1
PRESET	0	0.
GUARD	0	1

The frequency of data generation shall be controlled by the clock. The generator shall provide a two line output with signals of opposite phase. The amplitudes of the signals shall be nominally -8V and +2V. The output impedance shall be nominally 300 ohms.

The data generator shall be controlled by the FREQUENCY switches S18 through S21, the MODE switch, S16, and the PRESET CHAN switch, S17. In the GUARD mode the frequency shall be controlled by a fixed diode matrix in the Test Set providing a frequency of 243.000 MHz. The data generator shall be enabled under the following switch conditions: When the TESTED UNIT switch, S12, is in the RMT RT and RMT IND positions, when S12 is in the RMT CONT and RMT RT/RMT CONT positions and the TAKE CMD switch, S13 is in the RADIO position, and when S12 is in the PNL RAD position and S13 is in the RMT CONT position. The data generator shall be disabled when the GUARD PRECED switch, S5, is in the ON position.

3.7.10.1.5 Data receiver. The data receiver shall process the frequency control data generated wither by the Test Set or the unit under test and drive the FREQUENCY display.

3.7.10.2 Mount internal function assemblies. The Mount shall contain the following internal functional capabilities. Each functional capability shall not be required to be a separate assembly.

3.7.10.2.1 Blower. The Mount shall contain a blower system capable of properly cooling the unit under test during the performance of testing. Blower, B1, shall conform to MIL-B-23071.

3.7.10.2.2 Blower sensor. A sensor circuit shall be in the Mount to sense the airflow the Blower, B1, creates when it is operating. If this airflow is not sufficient the primary power being supplied to the unit under test will be interrupted.

3.7.11 Electrical power source. The test equipment shall operate from a power source as specified in MIL-T-28800. The power required shall be of the following type and shall not exceed the specified amount.

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DC Power, 28V, 10 amp.

AC Power, 115VAC±5 VAC, 60 Hz, 1 amp single phase

3.7.11.1 Protection. The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704.

3.7.12 Warm-up time. No warm up time shall be required prior to operation.

3.8 Dimensions. The maximum overall size of the AN/ARM-165A shall not exceed 20.0 inches wide by 15.0 inches high by 18.0 inches deep not including the pendant power cable or cable assemblies. The maximum overall size of the MT-4895/ARM-165A shall not exceed 24.0 inches wide by 9.0 inches high by 12.5 inches deep not including pendant cables.

3.9 Finish. Finishes shall be in accordance with MIL-T-28800.

3.10 Interchangeability. Interchangeability shall be in accordance with MIL-T-28800.

3.11 Workmanship. Workmanship shall be in accordance with MIL-T-28800.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

1. First article (preproduction) tests

2. Initial production tests

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3. Acceptance tests

4. Life test

4.4 First article (preproduction) tests. First article (preproduction) tests shall be made on an equipment representative of the production equipments to be supplied under the contract. The tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.4.1 First article (preproduction) test data. When specified in the contract first article test data shall be supplied in accordance with UDI-I-21349.

4.3 Scope of tests. The tests shall include all tests deemed necessary, by the procuring activity, to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Within the limits specified in MIL-T-28800 first article tests shall include environmental tests in accordance with MIL-T-5422 and interference test in accordance with MIL-STD-462 to the MIL-STD-461 requirements.

4.4.2 Maintainability assurance test. A maintainability insurance test approved by the procuring activity shall be conducted in accordance with MIL-STD-471 to demonstrate the requirements of 3.4.2. The test details shall include as a minimum, task selection, test method, demonstration support material, demonstration conditions, demonstration test team, and test schedule.

4.4.3 Probability of successful performance test. A probability of successful performance assurance test to demonstrate the requirement of 3.4.3 shall be conducted. The test procedure shall be prepared in accordance with 4.6 and submitted by the contractor to the procuring activity for approval.

4.4.4 First article (preproduction) approval. Approval of the test sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the test sample. Prefabrication of production equipment prior to the approval of the test sample is at the contractor's own risk. The approved test sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The test sample shall not be considered as one of the equipments under the contract.

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4.5 Production equipment. Equipment supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved test sample. Each equipment shall be capable of successfully passing the same tests as imposed on the test sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.6 Initial production tests. One of the first ten production equipment shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The test sample shall not be selected for this test.

4.6.1 Scope of tests. This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.6.2 Accessory material. In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.6.3 Initial production sample approval. Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.6.4 Reconditioning of initial production test sample. On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.7 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the govern-

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ment inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptable tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following.

(1)	Individual tests	(4.4.1)
(2)	Sampling tests	(4.4.2)
(3)	Reliability assurance tests	(4.4.3)
(4)	Special tests	(4.4.4)

4.7.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1)	Examination of product	(4.4.1.1)
(2)	Operational test	(4.4.1.2)
(3)	Manufacturing run in test	(4.4.1.3)

4.7.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.7.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. 1/

4.7.1.3 Manufacturing run in test. Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

1/ Approval of the operational test procedure must be obtained from the Procuring activity prior to testing.

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Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operating within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain that the failure was not a result of the vibration. Should repetitive failures occur, corrective actions shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.7.1.4 Environmental tests

4.7.1.4.1 Altitude. The equipment shall be tested in accordance with MIL-T-28800, Class 4 equipment.

4.7.1.4.2 Shock. The equipment shall be tested in accordance with MIL-T-28800, Class 4 equipment.

4.7.1.4.3 Vibration. The equipment shall be tested in accordance with MIL-T-28800, Class 4 equipment.

4.7.1.4.4 Humidity. The equipment shall be tested in accordance with MIL-T-28800, Class 4 equipment.

4.7.2 Environmental tests. Equipment selected for sampling shall first have passed the individual tests. Equipments shall be selected for sampling tests by the goverment inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action).

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Quantity of Equipment Offered for acceptance	Quantity to be Selected for Sampling Tests
First 10	0 2/
Next 50	1
Next 75	1
Next 100	1
One for each additional 200 or fraction thereof.	

4.7.2.1 Scope of tests. As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, marking all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restrictions on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.7.3 Reliability assurance tests. Reliability assurance tests shall be conducted using MIL-STD-781. Tests shall be conducted as required by both the Qualification Phase and Sampling Phase.

4.7.3.1 Qualification phase. Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The

2/ If by contract action the Initial production test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

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maximum number of equipment to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan VI shall be used.

4.7.3.2 Reliability sampling phase tests. Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase Test Level E shall be used. The Accept-Reject criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.7.3.2.1 Lot size for sampling phase. The equipments constructed during one month shall be one lot.

4.7.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.7.4 Special tests. Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.7.4.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.

4.7.4.2 Scope of tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.7.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

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(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity, for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.8 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test sample shall be performed under the conditions specified in 4.5.1. The life test shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests). When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipment equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipment Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
1 for each additional 500 or fraction thereof	

4.8.1 Test conditions. The life test shall be conducted under the following simulated service conditions:

Temperature	Room ambient
Altitude	Normal ground (0-5000 ft.)

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Humidity	Room ambient
AC Voltage	115 ±5 volts (at lowest applicable frequency)
DC Voltage	28.0 ±2.0 volts
Vibration	None

4.8.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.8.3 Performance check. At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.8.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failure, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

Test data shall be supplied in accordance with UDI-R-21139.

4.8.4.1 Failure report. In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall proposed suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable. Data shall be supplied in accordance with UDI-R-21140.

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4.8.5 Reconditioning of life test samples. An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.9 Test procedures. The procedures for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.10 Reconditioning of tested equipment. Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.11 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.12 Rejection and retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-

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17555, the method of preservation for Level A shall be determined in accordance with the selection chart of Appendix D of MIL-STD-794.

5.2 Marking. In addition to any special marking by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The equipment shall be for intermediate and depot level of maintenance and shall be capable of performing functional tests and fault isolation procedures on the SA-1964/ARC-159(V) Switching Unit.

6.2 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-28800, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Nonrepairable subassemblies. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplied), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.4.1 Title number and date of this specification

6.4.2 Selection of applicable levels of packaging and packing (See 5.1).

6.4.3 Specify first article test data in accordance with UDI-T-21349, Report, First Article (preproduction) Test, on data requirement list DD 1423 as required by paragraph 4.2.1.

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6.5 Type designation. The parentheses (*), when used in this type designation, will be deleated or replaced by either a number of letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.6. The complete type number shall be used on nameplates shipping records and instruction books, as applicable.

6.6 Associated equipment. The equipment shall operate with the following associated equipment or their equivalent.

Item Name	Federal Mfg. code	Manufacturers Part Number
Signal Generator	88869	FM2748
Power Meter	24655	ME-2A
Multimeter VTVM	80058	AN/USM-116C
RF Power Meter	80058	AN/URN-43B
GP Oscilloscope	80058	AN/USM-281A
Distortion Analyzer	28480	334A
Attenuator	80058	UG-528/U
Double Balanced Mixer	28480	10514A
Variable Atten.	24655	874GAL
Crystal Detector	28480	420B
Multimeter, VOM	80058	AN/USM-311
Digital Voltmeter	89536	8300A01-02
Decade Resistor	24655	1433-9714 Type 1433M
RF Voltmeter	04901	91H-S7
Sweep Sig. Gen.	80058	SG-1020/UR
Spectrum Analyzer	80009	491
Pulse Generator	06811	110B

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Item Name	Federal Mfg. Code	Manufacturer's Part Number
Frequency Counter	28480	H61-5245L
Frequency Counter Converter	28480	5253B
Audio Oscillator	28480	200CD
Translator Test Set	13499	MX-9626/ARM

6.7 Definitions

Abbreviations	Definitions
J	Jack
W	Cable
P	Plug
MT	Mount
DC	Direct Current
AC	Alternate Current
RF	Radio Frequency
AGC	Automatic Gain Control
REXMT	Retransmit
CONT	Control
IF	Intermediate Frequency
M	Meter
V	Volt
Hz	Hertz
PWR	Power

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Abbreviation	Definition
UHF	Ultra High Frequency
CB	Circuit Breaker
S	Switch
DS	Lamp
RMT	Remote
IND	Indicator
EQPT	Equipment
R/T, RT	Receiver-Transmitter
SIG	Signal
GEN	Generator
X-MODE	Auxiliary Mode
XMT	Transmit
CASS	Command Activated Sobobouy System
UNATTEN	Unattenuated
MIC	Microphone
CW	Continuous Wave
RCVD	Received
ADF	Automatic Direction Field
IMPD	Impedance
PRECED	Precedance
ANT	Antenna
SEL	Select

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Abbreviation	Definition
FREQ	Frequency
PNL	Panel
RAD	Radio
CMD	Command
CHAN	Channel
BW	Bandwidth
RCVR	Receiver
PTT	Push-to-Talk
INTRPT	Interrupt

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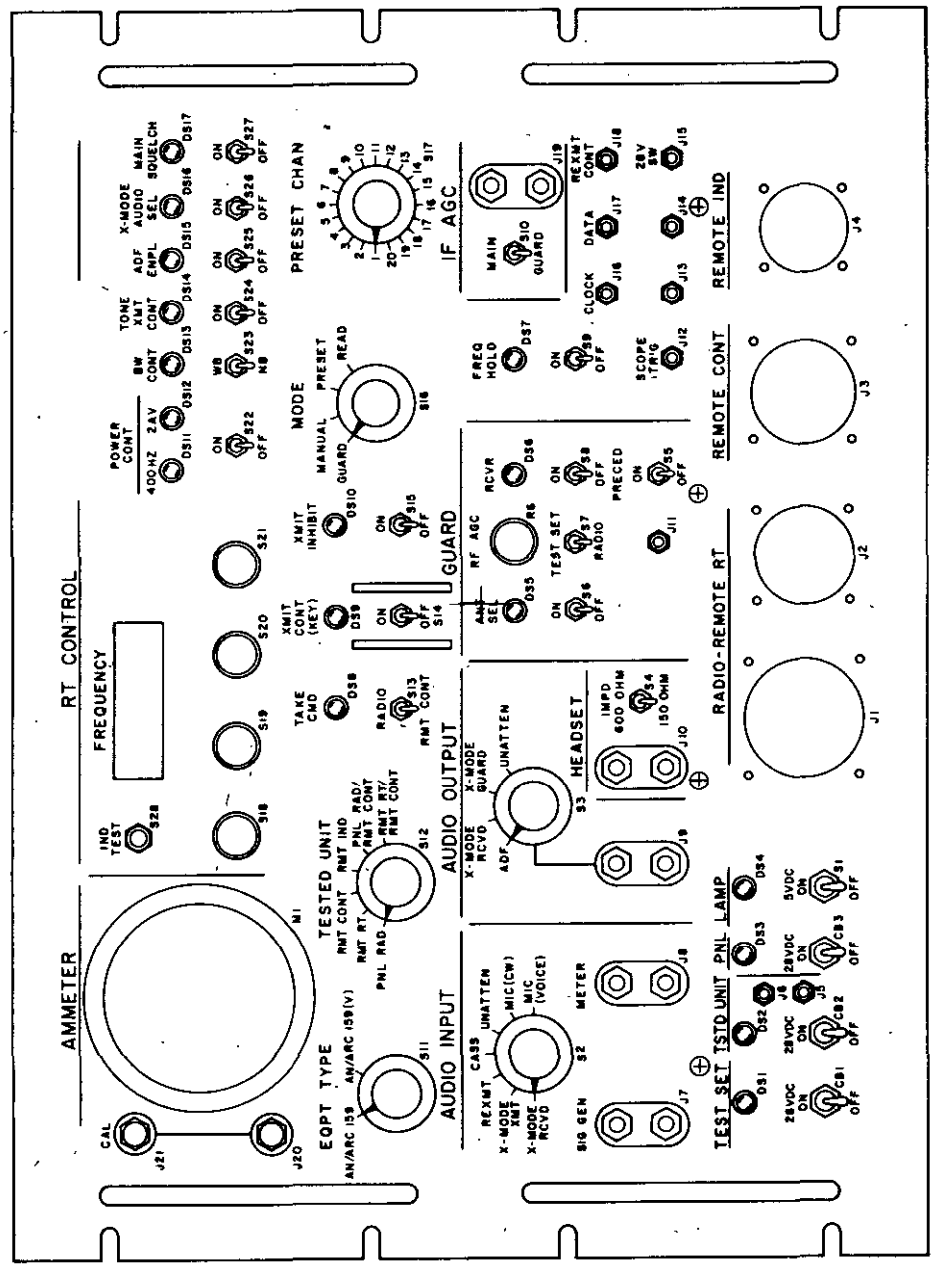
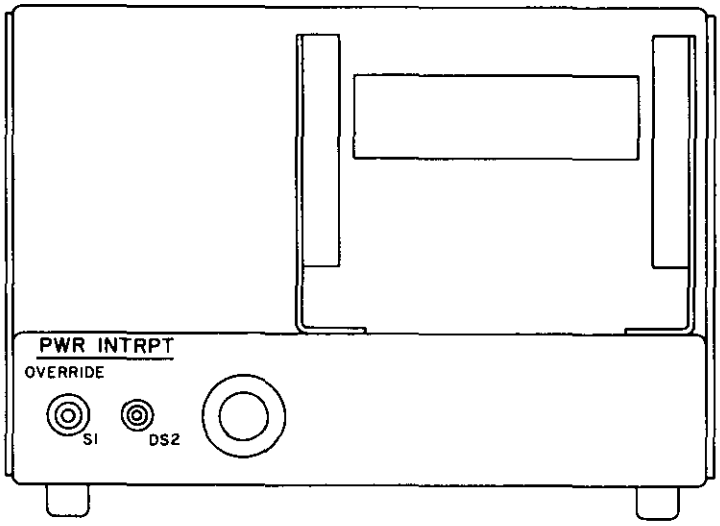
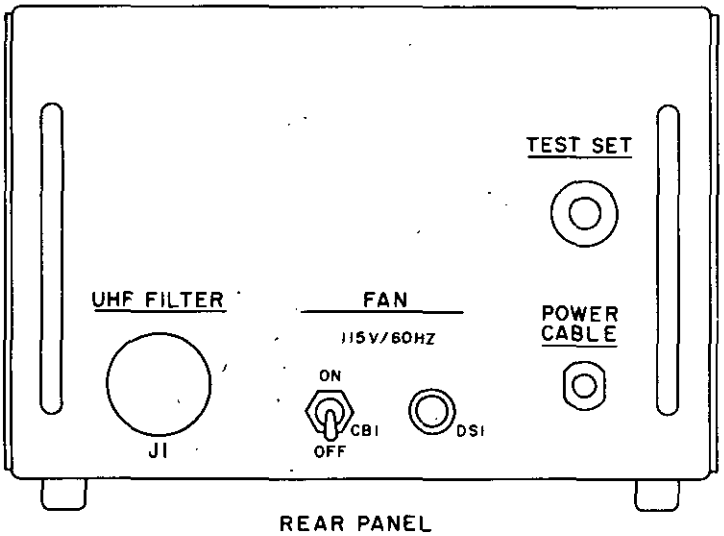


FIGURE 1
TEST SET FRONT PANEL

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FRONT PANEL



REAR PANEL

FIGURE 2
MT-4895/ARM-165A
FRONT AND REAR PANEL

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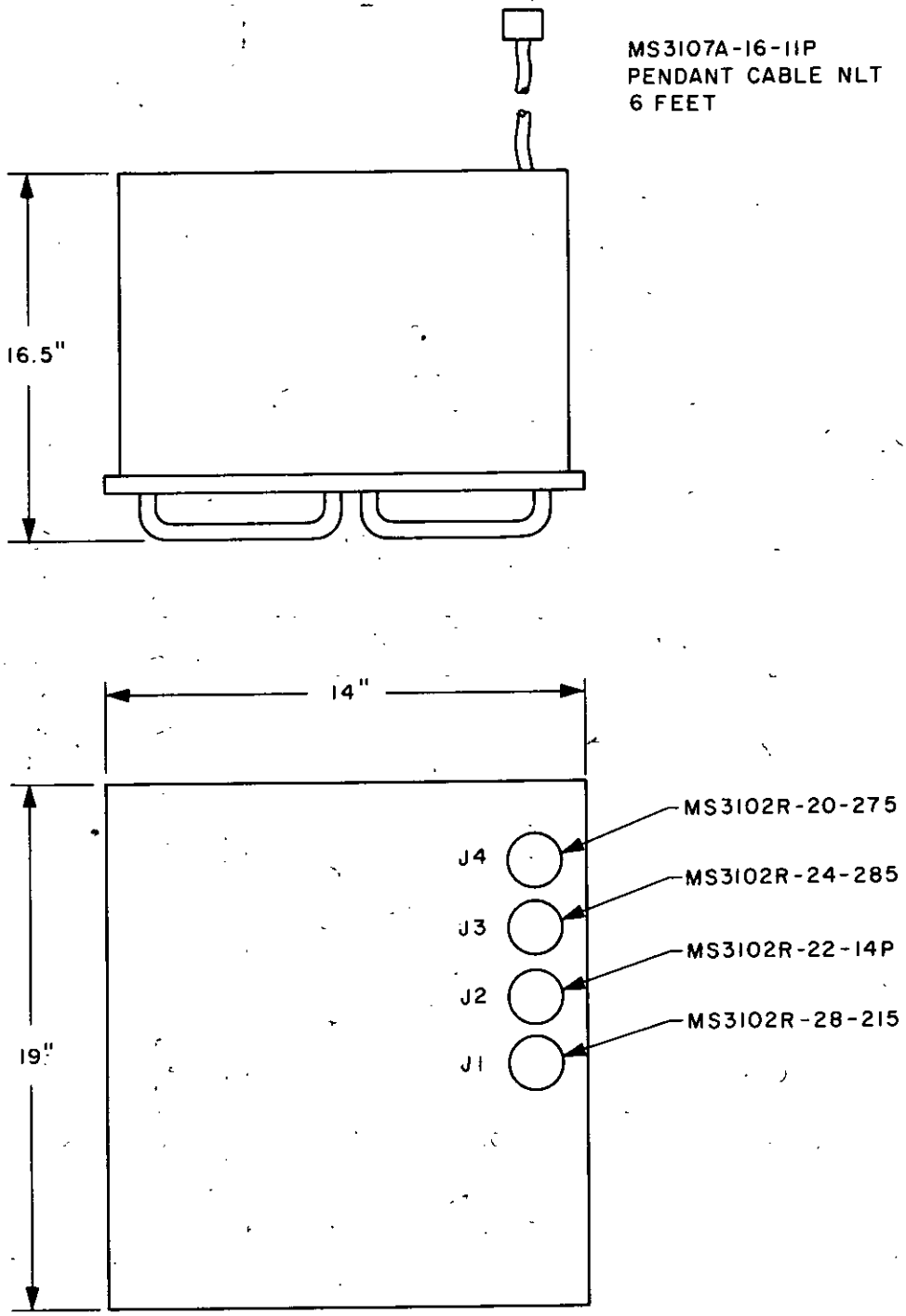


FIGURE 3

OUTLINE DIMENSIONS OF TEST SET PANEL

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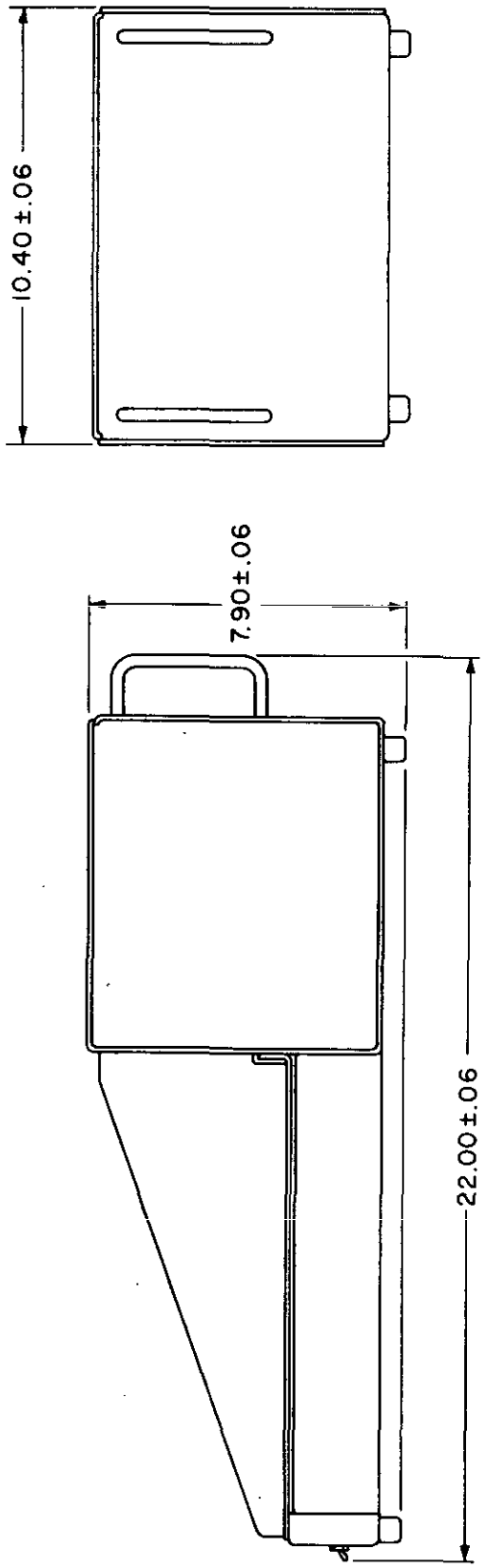


FIGURE 4
OUTLINE DIMENSIONS OF MT-4895/ARM-165A

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